

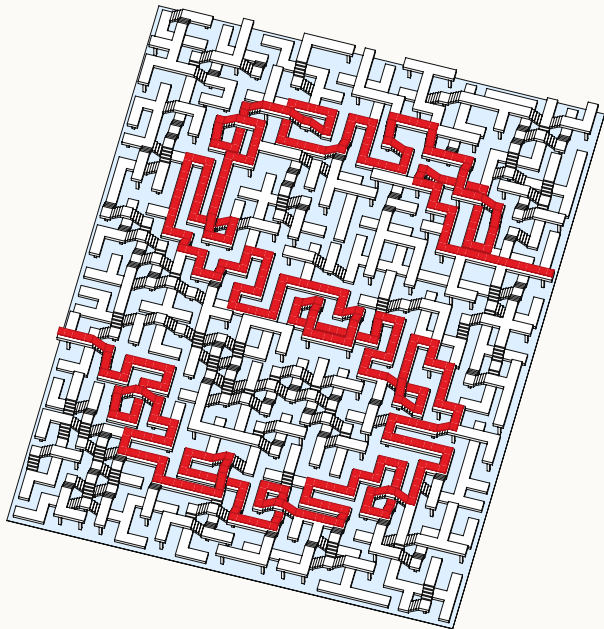
From Patterns to Maze Solutions

An SMT-Based Construction

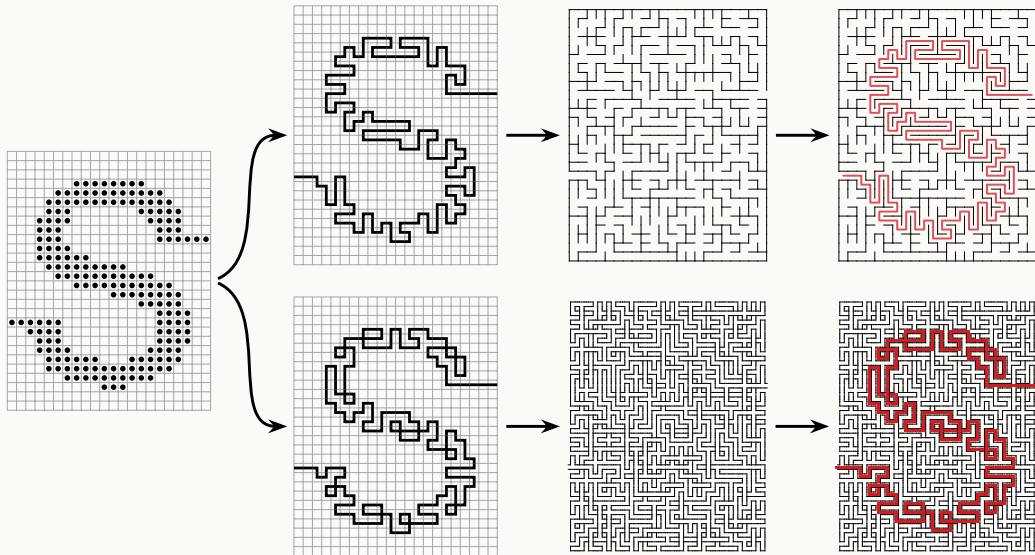
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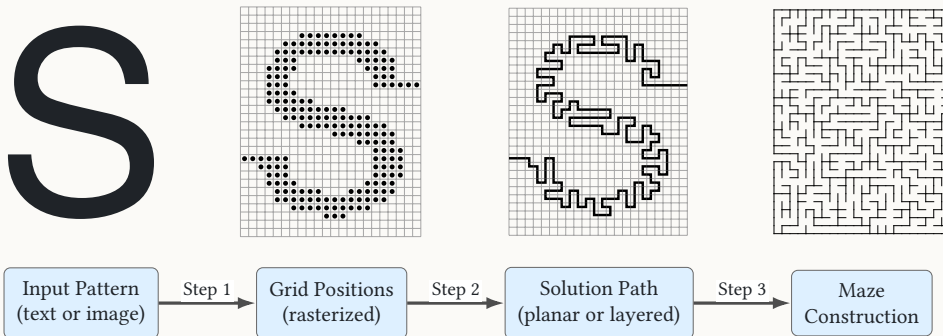
Bridges 2026



Visual Motivation



Pipeline Overview



1. Grid positions

Rasterize text or shape into a finite set $P \subset \mathbb{N}^2$.

2. Solution path

Use SMT to synthesize a planar or layered route.

3. Maze construction

Complete the route into a perfect maze.

The Path-Synthesis Problem

Rasterization gives a finite target region and two prescribed endpoints.

$$\begin{aligned}P &\subset \mathbb{N}^2, & s, t &\in P \\s &= a_0, a_1, \dots, a_n = t, & a_i &\in P, \\ \|a_{i+1} - a_i\|_1 &= 1.\end{aligned}$$

Among all valid paths, maximize n ; in the planar case, full coverage gives $n + 1 = |P|$.

Planar

Each grid position is visited at most once.

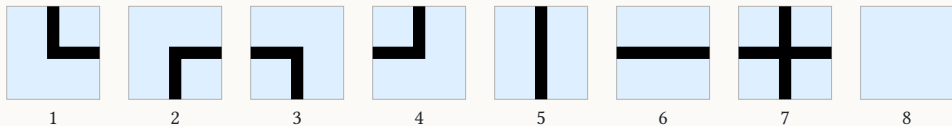
Woven

A crossing permits horizontal and vertical visits at one position.

Coverage

Occupy as many target positions as the path constraints allow.

Tile-Based Path Representation



- For each $(r, c) \in P$, introduce a tile variable $t_{r,c} \in \{1, \dots, 8\}$.

- Planar paths are the special case where crossing tiles are forbidden.

$$\text{inRange}(t) = 1 \leq t \leq 8,$$

$$\text{hasUp}(t) = t \in \{1, 4, 5, 7\},$$

$$\text{hasRight}(t) = t \in \{1, 2, 6, 7\},$$

$$\text{nonWhite}(t) = \mathbf{1}_{t \neq 8},$$

$$\text{hasDown}(t) = t \in \{2, 3, 5, 7\},$$

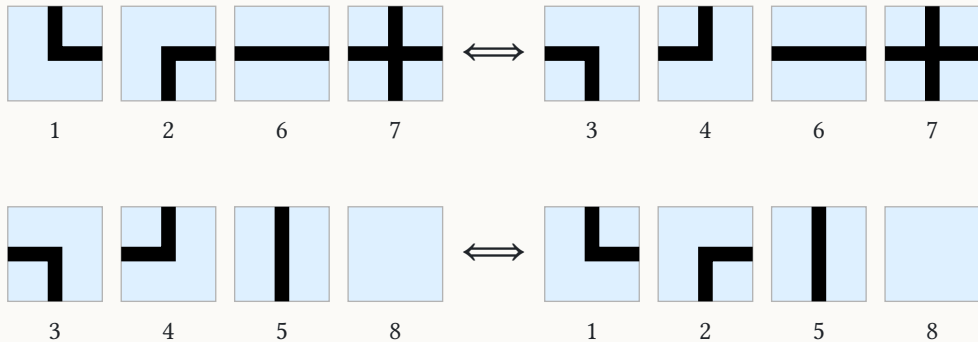
$$\text{hasLeft}(t) = t \in \{3, 4, 6, 7\}.$$

$$C = \sum_{(r,c) \in P} \text{nonWhite}(t_{r,c}),$$

$$C \geq K \quad \text{or maximize } C.$$

Local Rules Build a Legal Grid Path

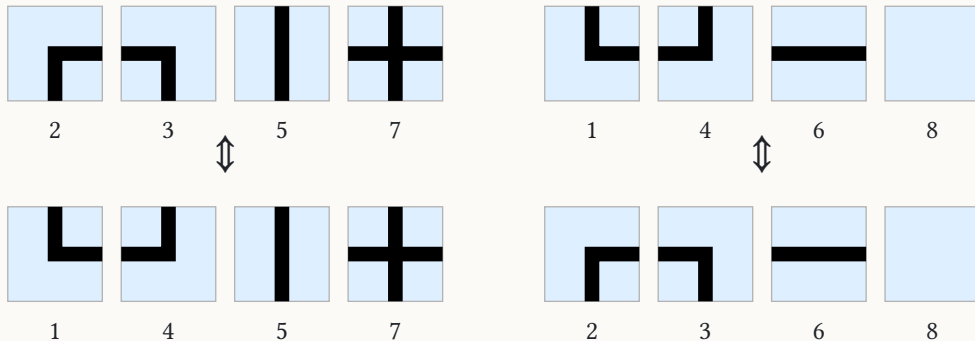
Horizontal neighbors must agree on their shared edge



$$\text{hasRight}(t_{r,c}) \iff \text{hasLeft}(t_{r,c+1})$$

Local Rules Build a Legal Grid Path

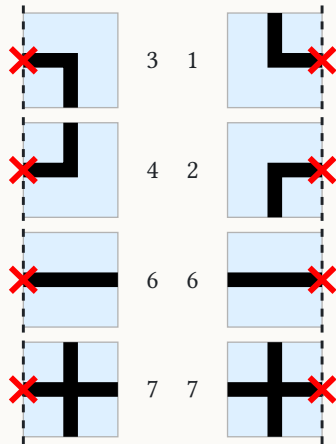
Vertical neighbors must agree on their shared edge



$$\text{hasDown}(t_{r,c}) \iff \text{hasUp}(t_{r+1,c})$$

Local Rules Build a Legal Grid Path

Boundary constraints



- ▶ For each non-endpoint position $(r, c) \in P \setminus \{s, t\}$, a missing neighbor disables the corresponding exit:

$$(r, c - 1) \notin P \implies \neg \text{hasLeft}(t_{r,c}),$$

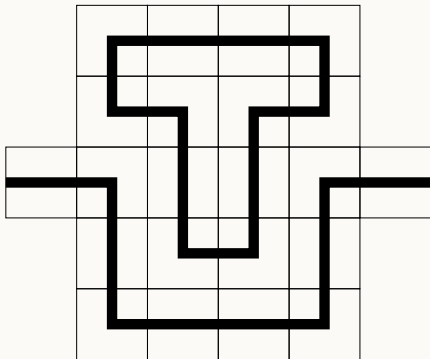
$$(r, c + 1) \notin P \implies \neg \text{hasRight}(t_{r,c}),$$

$$(r - 1, c) \notin P \implies \neg \text{hasUp}(t_{r,c}),$$

$$(r + 1, c) \notin P \implies \neg \text{hasDown}(t_{r,c}).$$

- ▶ The prescribed endpoints s and t are the only exceptions to these boundary constraints.
- ▶ At each exception, the boundary side fixes a straight tile: horizontal for left/right, and vertical for top/bottom.

Local Constraints Are Not Enough

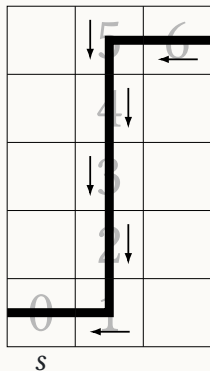


Local compatibility does not imply global connectivity.

- ▶ The endpoint-to-endpoint route satisfies every adjacency and boundary rule.
- ▶ So does the detached loop: no local constraint detects that it is disconnected.

We need a global connectivity certificate.

Global Connectivity: Parent and Rank



For each occupied position (r, c) , introduce

$$\rho_{r,c} \in \mathbb{Z}, \quad p_{r,c}^d \in \text{Bool}, \quad d \in \{R, L, U, D\}.$$

Here $\rho_{r,c}$ is its rank; $p_{r,c}^d$ records whether direction d is selected as a predecessor.

- ▶ Root s : $\rho_s = 0$, and no predecessor is selected.
- ▶ Every other occupied position chooses at least one predecessor and has rank in $\{1, \dots, |P|\}$:

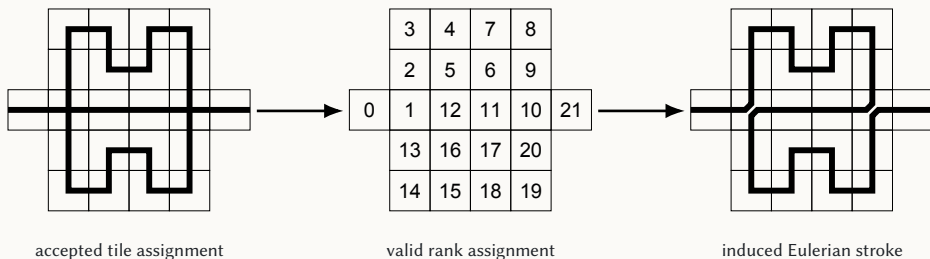
$$\text{nonWhite}(t_{r,c}) = 1 \wedge (r, c) \neq s \implies \left(\bigvee_d p_{r,c}^d \right) \wedge 1 \leq \rho_{r,c} \leq |P|.$$

A predecessor must follow an existing local path edge and have smaller rank. For example,

$$\rho_v > \rho_{\text{parent}(v)} > \dots > \rho_s = 0$$

$$p_{r,c}^R \implies \text{hasRight}(t_{r,c}) \wedge \text{hasLeft}(t_{r,c+1}) \wedge \rho_{r,c+1} < \rho_{r,c}.$$

One Rank per Grid Position Is Still Not Enough



- ▶ One rank per grid position merges the horizontal and vertical strands of a crossing into a single graph vertex.
- ▶ The model can therefore certify a single-stroke Eulerian traversal, but not the intended woven over-under path.

Crossing Tiles as Virtual Path Elements

A tile activates one or two path elements:

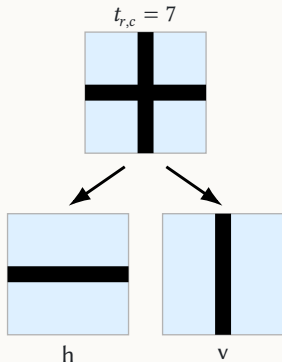
$$A_{r,c}^{\text{base}} = (\text{nonWhite}(t_{r,c}) = 1) \wedge (t_{r,c} \neq 7),$$

$$A_{r,c}^{\text{h}} = A_{r,c}^{\text{v}} = (t_{r,c} = 7).$$

Each $k \in \{\text{base}, \text{h}, \text{v}\}$ gets its own parent-rank state:

$$\rho_{r,c}^k \in \mathbb{Z}, \quad p_{r,c}^{d,k} \in \text{Bool},$$

$$A_{r,c}^k \wedge (r,c) \neq s \implies \left(\bigvee_d p_{r,c}^{d,k} \right) \wedge 1 \leq \rho_{r,c}^k \leq |P|.$$



- ▶ The h and v elements permit only horizontal and vertical parent directions, respectively.
- ▶ Coverage remains based on $\text{nonWhite}(t_{r,c})$, so a crossing is counted once.

SMT-LIB Realism

A directional parent constraint and its concrete encoding

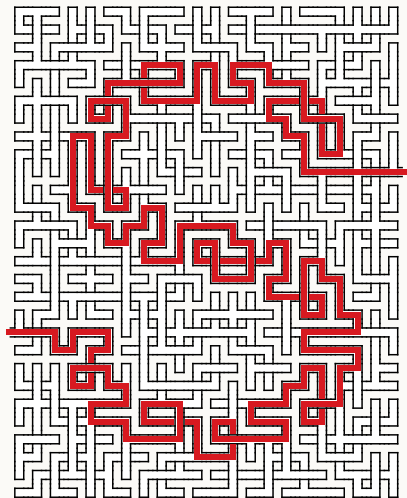
$$p_{r,c}^{R,h} \Rightarrow (A_{r,c}^h \wedge \text{hasRight}(t_{r,c})) \wedge \text{hasLeft}(t_{r,c+1}) \wedge (A_{r,c+1}^{\text{base}} \wedge \rho_{r,c+1}^{\text{base}} < \rho_{r,c}^h \vee A_{r,c+1}^h \wedge \rho_{r,c+1}^h < \rho_{r,c}^h).$$

In the SMT encoding, this implication expands into a disjunction over the possible kinds of the neighboring tile; for instance, with $(r, c) = (31, 13)$:

```
(assert (=> pR_31_13_h
  (or (and (= tile_31_14 7)
    (= tile_31_13 7) (hasRight tile_31_13) (hasLeft tile_31_14)
    (< rank_31_14_h rank_31_13_h))
    (and (and (not (= tile_31_14 7)) (= (nonWhite tile_31_14) 1))
      (= tile_31_13 7) (hasRight tile_31_13) (hasLeft tile_31_14)
      (< rank_31_14_base rank_31_13_h))))))
```

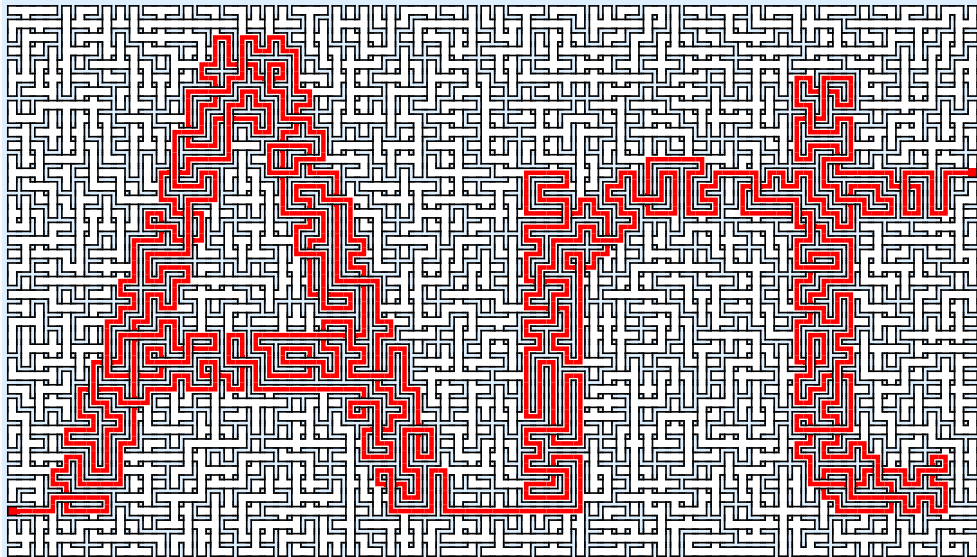
- ▶ The constraints are emitted as SMT-LIB, not pseudocode.
- ▶ This is the horizontal parent implication for a crossing virtual element.
- ▶ The neighbor can be another crossing h element or an ordinary base element.

Maze Completion from a Prescribed Solution Path

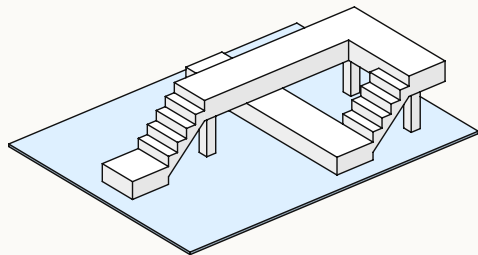


- ▶ Read the SMT model: each value $t_{r,c}$ selects a tile, and the tile assignment draws the prescribed solution path.
- ▶ Planar: insert the solution-path edges first, then attach each remaining grid position by a Wilson-style loop-erased random walk that stops at the existing tree.
- ▶ Woven: insert required crossings and solution links first, then use Kruskal-style completion on the expanded over/under graph.
- ▶ The completed passage graph is a tree, so the entry–exit solution is unique.
- ▶ Render the passage graph using standard walls and passages, with inset corridors used to depict woven crossings.

A Big Example

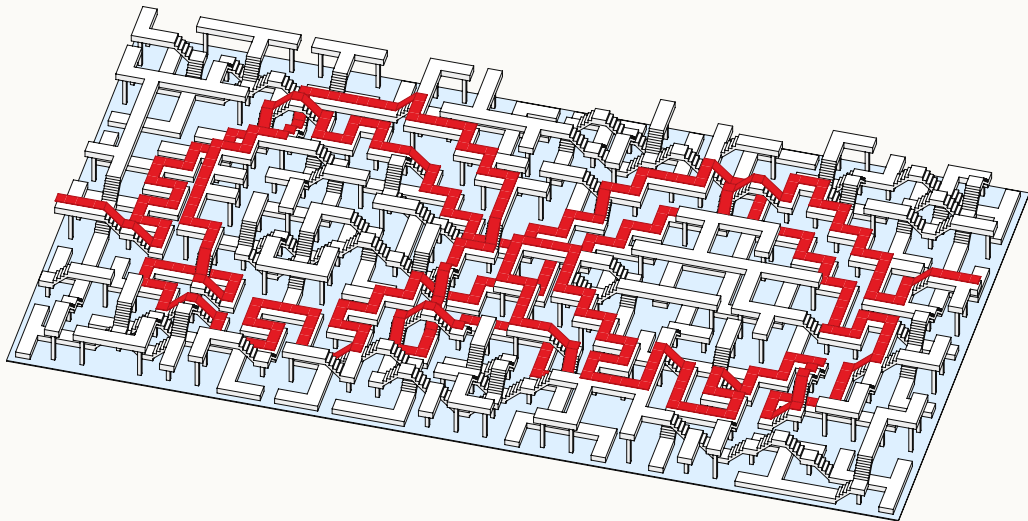


After the Core Construction: 3D Realization



- ▶ The SMT construction already ends with a valid combinatorial path.
- ▶ A separate visualization stage can realize the woven topology with local 3D primitives:
 - ▶ platforms and bridges,
 - ▶ connectors and stairs,
 - ▶ support posts, openings, and a solution ribbon.
- ▶ This geometric layer displays the over-under relation; it does not alter the synthesized path or maze graph.

A 3D View of the Completed Woven Maze



Takeaways

Model the pattern

Target pixels, endpoints, and coverage define the path-synthesis problem.

Guarantee one path

Local tile rules combine with parent-rank connectivity.

Support two topologies

Virtual crossing elements specialize one encoding to planar or woven paths.

SMT synthesizes the prescribed solution path; standard spanning-tree methods then complete it into a perfect maze.

The extended report provides fuller SMT-LIB examples and details of maze completion and optional 3D realization.



Extended Report

arXiv:2607.09781